XGBoost Model

1. Model Overview

XGBoost (Extreme Gradient Boosting) is an ensemble model that builds many decision trees one after another, where each tree tries to fix the mistakes made by the previous one. It is well-suited for tabular data and handles class imbalance reasonably well.

2. Hyperparameter Tuning

A manual grid search was performed over the following values to find the best combination:

- max_depth: 2, 4, 6 — controls how deep each tree grows.
- n_estimators: 100, 500, 700, 1000 — number of trees to build.
- learning_rate: 0.01, 0.05, 0.1 — how much each tree contributes to the final result.

All other hyperparameters were fixed: subsample = 0.7, colsample_bytree = 0.7, eval_metric = AUC, and random_state = 433. Each combination was trained on the training set and evaluated on the validation set. The best combination was the one with the highest validation accuracy.

3. Best Hyperparameters

- max_depth: 4
- n_estimators: 500
- learning_rate: 0.05

These values gave a validation accuracy of 87.78% and a train-to-val ratio of 0.993, which means the model generalized well without overfitting. Deeper trees (max_depth = 6) and more estimators showed signs of overfitting, where training accuracy kept rising but validation accuracy dropped.

4. Final Training

After selecting the best parameters, the model was retrained on the full training and validation data combined (about 26,000 rows) to make the most use of available data. It was then evaluated on the held-out test set.

5. Results on Test Set

- Accuracy: 86.91%
- AUC Score: 0.9270
- Class 0 (<=50K) — Precision: 89%, Recall: 94%, F1: 91%
- Class 1 (>50K) — Precision: 78%, Recall: 67%, F1: 72%